

OMS Digital Pandemic Preparedness workshop: Agenda

Venue: Oxford Martin School Lecture Theatre, 34 Broad Street, Oxford, OX1 3BD

Monday 21st September

10:00 Registration and coffee
10:30 Session 1: *Welcome and workshop overview*
12:30 Lunch
13:30 Session 2: *UKHSA-led session*
15:00 Break
15:30 Session 3: *UKHSA-led session*
17:00 End

Tuesday 22nd September

09:00 Session 4: *Global perspectives on potential and challenges for digital tools*
10:30 Break
11:00 Session 5: *Ethical, efficient and responsible decision-making*
12:30 Lunch
13:30 Session 6: *Future requirements and opportunities, motivated by past experiences*
15:00 Break
15:30 Session 7: *Global and pathogenic variation in requirements*
17:00 End

19:30 Dinner

Wednesday 23rd September

09:00 Session 8: *The potential of data, and the challenges of collecting and sharing it*
10:30 Break
11:00 Session 9: *Behavioural and social considerations, and ways to measure impacts*
12:30 Lunch
13:30 Session 10: *What next?*
15:00 End

OMS Digital Pandemic Preparedness workshop: Programme

Monday 21st September

Session 1: Welcome and workshop overview

- Welcome and general notices
- Christophe Fraser and the OMS Digital Pandemic Preparedness Team

The OMS Digital Pandemic Preparedness project background. Scope and aims for the three days and beyond.

Lightning talks: a rapid round-up of Programme work so far (each 2 slides, 5 mins max):

- Younjung Kim / Christl Donnelly **Federated learning from privacy-preserving data**
- Jamie Webb **A platform approach to ethics for digital pandemic preparedness**
- Michael Parker **Ethics of Digital Pandemic Preparedness**
- Thomas Rawson **Infectious disease and demography**
- Melinda Mills **Future work and Pand-AI**
- Luca Ferretti **Opportunities for epidemiological insights from app data**

12:30 *Lunch*

Session 2: UKHSA-led session

13:30 *Leanne Cummings Director of Products and Services for Government Digital Service*

14:15 *Steven Riley Director General and Chief Data Officer, UK Health Security Agency (UKHSA)*
Preparing for the Next Pandemic: Prioritising Digital Technology Interventions

15:00 *Break*

Session 3: UKHSA-led session

15:30 Break-out sessions led by Steven Riley

17:00 *End*

Tuesday 22nd September

Session 4: *Global perspectives on potential and challenges for digital tools*

Chair: Peter Horby

9:00 Randy Marsden *former project lead of Exposure Notifications at Apple*

The Triangle of Trust: Lessons from GAEN for the Next Pandemic

9:30 Jason Bay, *National University of Singapore; TraceTogether (Singapore)*

TraceTogether: Reflections on the future of Digital Contact Tracing Apps

10:00 Maria van Kerkhove *Director, Department of Epidemic and Pandemic Management, WHO Health Emergencies Programme*

Learnings and challenges from COVID-19, mpox, Ebola and more: how can digital public health tools help us to prevent and respond quicker for future events

10:30 *Break*

Session 5: *Ethical, efficient and responsible decision-making*

Chair: Michael Parker

11:00 Devi Sridhar *(online), Global Public Health, University of Edinburgh, UK*

A decision-making tree for policy responses to a pathogen with pandemic potential

11:30 Stephen John *Philosophy of Public Health, University of Cambridge, UK*

Pandemic policymaking, multiple perspectives and unexpected consequences: lessons from school closures

12:00 **Panel discussion:** *where does responsibility lie for a digital public health tool and its surrounding infrastructure? How can ethics be embedded? (Devi Sridhar, Stephen John, Peter Horby)*

12:30 *Lunch*

Session 6: *Future requirements and opportunities, motivated by past experiences*

Chair: Luca Ferretti

13:30 Wolfgang Emmerich Zuhlke (*NHS COVID-19 app*), *Computer Science, University College London, UK*

Providing a Digital Contact Tracing service in the first week of a Pandemic

14:00 Justus Benzler *Epidemiologist, Robert Koch Institute, Germany (Corona-Warn-App)*

Future Requirements and Opportunities for Universal and Frequent Testing - motivated by the German experience with the (imperfect) integration of lab- and community-based testing into an exposure notification app infrastructure

14:20 Vittoria Colizza *Founding Director of the Sorbonne Université Modeling Outbreaks Center; Director of Research at INSERM and Sorbonne Université, Paris, France*

From mobility traces to epidemic-relevant behaviour: lessons for pandemic preparedness

14:40 Samir Bhatt *Machine Learning and Public Health, University of Copenhagen, Denmark and Statistics and Public Health, Imperial College London, UK*

AI in Infectious Disease: An overview

15:00 *Break*

Session 7: *Global and pathogenic variation in requirements*

Chair: Christophe Fraser

15:30 Njenga Kariuki (*online*), *Global Health, Washington State University, USA*
 (suggested) perspectives for the global South, challenges and opportunities for digital tools for LMICs

16:00 Katherine Littler, *Global Health ethics and governance, WHO Geneva, Switzerland*
 (suggested) international governance: how to prepare digital public health tools to be crisis-ready with international interoperability while respecting the ethical requirements and responsibilities of individual public health agencies and technology platforms

16:30 **Panel discussion:** *variation by country, setting and pathogen – opportunities and challenges. How to build trust and create trustworthy digital tools?* (Njenga Kariuki, Katherine Littler, Jason Bay, Justus Benzler)

17:00 *End*

19:00 - 21:30 Dinner at Oxford University Museum of Natural History, Parks Road, OX1 3PW

Wednesday 23rd September

Session 8: *The potential of data, and the challenges of collecting and sharing it*

Chair: Christl Donnelly

9:00 Simon Cauchemez *Institut Pasteur, France*

(suggested) the potential and implications of using data from digital public health devices for epidemiological inference. Benefits and challenges, particularly when compared to conventional inference approaches.

9:20 Navjot Kalra *Director of CEPI Digital*

Intelligence for Value (as socio-technical framework) for practical considerations including people, process and technology for enabling sharing intelligence and insights, and how to have the right capabilities in place to be able to do so in a crisis.

9:40 James Wilson *philosophy, ethics and public health policy, University College London, UK*

Public health ethics, policy, and digital public health tools: how much, and what kinds of data should be collected? Exploring the trade-offs between maximising population health, ensuring equity, respecting individual privacy, and maintaining trust.

10:00 *Panel discussion: trade-offs in data collection and utility, safe data sharing, and the role of AI for digital public health tools (Simon Cauchemez, Navjot Kalra, James Wilson)*

10:30 *Break*

Session 9: *Behavioural and social considerations, and ways to measure impacts*

Chair: Melinda Mills

11:00 Ben Cowling *(online) Epidemiology and Public Health, Hong Kong University, Hong Kong*

Behavioral responses to public health measures, pandemic fatigue, and risk compensation

11:20 Helen Ward *Public Health, Imperial College London, UK*

The use of digital public health tools and remote testing in pandemic preparedness: lessons from the REACT study

11:40 Ciro Cattuto *ISI Foundation, Italy*

Measuring human social networks with wearable devices: lessons from SocioPatterns and the road ahead

12:00 *Panel discussion: how can behavioural and social considerations be dynamically captured and embedded in pandemic response? (Ben Cowling, Helen Ward, Ciro Cattuto)*

12:30 *Lunch*

Session 10: *What next?*

13:30 What next?

Chair: Michelle Kendall

Discussion; round-up and plan towards writing a community-led Blueprint for Digital Pandemic Preparedness. Should there be future annual workshops?

15:00 *End*